**1** **(a)** One end of a cable is connected to a fixed point. A load is attached to the other end so that the wire hangs vertically.

The initial unstretched length $L$, the diameter $d$ of the cable, the applied load $F$ and the resulting extension $x$ are measured as

$L = 2.000 \pm 0.002$ m

$d = 0.40 \pm 0.02$ mm

$F = 25.0 \pm 0.5$ N

$x = 1.20 \pm 0.06$ mm

The Young modulus $E$ of the material of the cable is calculated by using the expression

$$E = \frac{4FL}{\pi d^2 x}$$

Calculate $E$ and the uncertainty in its value.

Show how your answers were obtained.

$E$ = ..... ± ..... N m$^{-2}$ [4]

**(b)** A uniform rectangular sign of length 0.80 m and width 0.68 m is fixed at its corner points A and B to a wall. The sign is also supported by a wire CD, as shown in Fig. 1.1

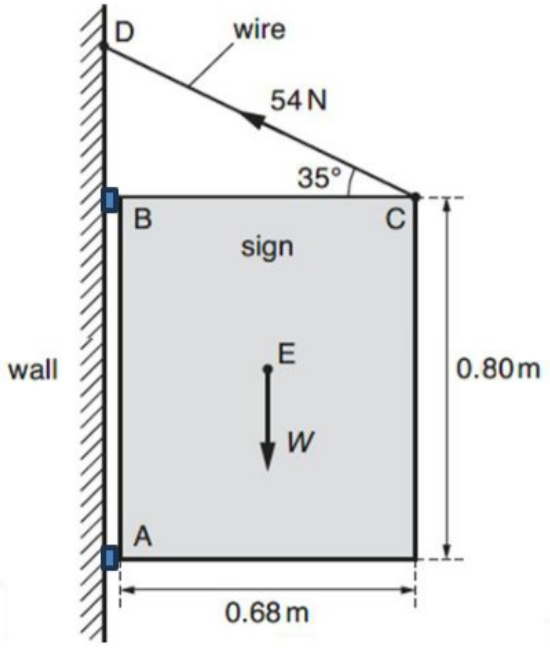



**Fig. 1.1**

The sign has weight *W* and centre of gravity at point E. The sign is held in a vertical plane with side BC horizontal. The wire is at an angle of 35° to side BC. The tension in the wire is 54 N.

The force exerted on the sign at B is only in the vertical direction.

**(i)** State two conditions necessary for the sign to be in equilibrium.

.....

.....

.....

..... [2]

**(ii)** Explain why the force on the sign at B does not have any moment about point A.

.....

..... [1]

**(iii)** Determine the weight *W* of the sign.

*W* = ..... N [2]

**(iv)** Calculate the total vertical force exerted by the wall on the sign at points A and B.

total vertical force = ..... N [2]

**(v)** The attachment at corner point B is removed but the sign remains stationary, supported solely by the wire CD and its remaining contact with the wall at corner point A.

On Fig. 1.1, draw an arrow to show the direction of the force exerted by the wall on the sign at point A. [1]

[Total: 12]